

Practical - 29

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**29) Create a simple Spark MLlib pipeline to classify data.
Use logistic regression or decision trees.**

```
import org.apache.spark.sql.SparkSession
import org.apache.spark.ml.feature.VectorAssembler
import org.apache.spark.ml.classification.LogisticRegression

object scala1 extends App {

  val spark = SparkSession.builder()
    .appName("Classification")
    .master("local[*]")
    .getOrCreate()

  import spark.implicits._

  val data = Seq(
    (1.0, 0.0),
    (2.0, 0.0),
    (3.0, 0.0),
    (7.0, 1.0),
    (8.0, 1.0),
    (9.0, 1.0)
  ).toDF("marks", "label")
```

```

val a = new VectorAssembler()
  .setInputCols(Array("marks"))
  .setOutputCol("features")

val d = a.transform(data)

val model = new LogisticRegression()
  .setLabelCol("label")
  .setFeaturesCol("features")

val result = model.fit(d)

val prediction = result.transform(d)

prediction.select("marks", "label", "prediction").show()

spark.stop()
}

```

Output.

```

+-----+-----+-----+
|marks|label|prediction|
+-----+-----+-----+
|  1.0|  0.0|        0.0|
|  2.0|  0.0|        0.0|
|  3.0|  0.0|        0.0|
|  7.0|  1.0|        1.0|
|  8.0|  1.0|        1.0|
|  9.0|  1.0|        1.0|
+-----+-----+-----+

```